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LIME vs SHAP: A Comprehensive Empirical Comparison

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Abstract

LIME and SHAP are widely used explainability methods, yet guidance on when to use each remains limited and often anecdotal. We present the first comprehensive empirical comparison evaluating both methods across five primary technical metrics with a secondary semantic analysis layer, using a rigorous benchmarking framework. Our study on tabular data reveals strong, statistically robust directional separation: SHAP dominates on stability and faithfulness-oriented quality metrics, while LIME yields substantially sparser explanations and lower latency in most matched contexts. The cost gap is large but heterogeneous across model families, with a heavy-tailed SHAP runtime distribution (matched-cell medians: 1,098.1 ms for SHAP versus 64.4 ms for LIME; means: 503,870.2 ms versus 3,083.1 ms). Based on these findings, we provide deployment-oriented guidance: SHAP for high-assurance auditing and LIME for strict latency-oriented interfaces.

Keywords: explainable ai, lime, shap, interpretability, robustness benchmarking

1 Introduction

Selecting an explanation method in production can be posed as a constrained decision problem: given a fixed predictor f and instance x, choose $E \in \{\text{LIME}, \text{SHAP}\}$ to maximize explanation quality while satisfying operational constraints. Concretely, we consider

$$E^\star = \arg \max_{E \in \{\text{LIME,SHAP}\}} U(S_E, F_E, \Delta_{k,E}, P_E) \quad \text{s.t.} \quad C_E \leq \tau,$$

where S is stability, F fidelity, Δ_k faithfulness gap, P sparsity/parsimony, C cost, and τ a latency budget.

LIME and SHAP optimize different principles. LIME explains locally by fitting a sparse surrogate around x,

$$\xi(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g),$$

favoring local fidelity and explicit sparsity controls (Ribeiro et al., 2016). SHAP explains predictions through additive feature attributions $f(x) = \phi_0 + \sum_i \phi_i$, where ϕ_i are Shapley-based contributions with game-theoretic guarantees (Lundberg and Lee, 2017). These theoretical differences are well known; however, they do not by themselves resolve deployment choices across model families, sampling regimes, and runtime constraints (Retzlaff et al., 2024; Ali et al., 2023).

Recent synthesis and systematic studies show that XAI evaluation remains fragmented, with no single accepted protocol and strong dependence on which evaluation dimensions are

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prioritized (Longo et al., 2024; Ali et al., 2023; Pawlicki et al., 2024; Canha et al., 2025). This makes single-metric comparisons insufficient for claim-ready method selection.

Open benchmarking toolchains and robustness-aware faithfulness protocols further show that method rankings can change under different metric families and perturbation settings (Agarwal et al., 2022; Sithakoul et al., 2024; Zheng et al., 2025).

This paper provides a paired, multi-objective benchmark using an audited, configuration-driven protocol on tabular data (UCI Adult). We evaluate five technical endpoints with explicit directionality: Stability (S, higher is better), Fidelity (F, higher is better), Faithfulness Gap (Δ_k, higher is better under top-k masking), Sparsity (active-feature ratio P, lower is better for parsimony), and Computational Cost (C, lower is better). Inference is performed on matched SHAP-LIME cells with identical $(\text{model}, \text{seed}, N)$, using paired Wilcoxon tests (Wilcoxon, 1945), paired effect sizes (d_z) (Lakens, 2013), and Holm correction (Holm, 1979).

Our paired analysis indicates a consistent trade-off: SHAP tends to provide stronger stability and fidelity signals, while LIME offers substantially lower runtime cost. This supports objective-driven method selection rather than universal method supremacy (Agrawal et al., 2025).

Our contributions are:

1. **Formalized deployment trade-off**: we frame LIME-vs-SHAP as a constrained multi-objective selection problem over quality, parsimony, and cost.

2. **Paired inferential evidence**: we quantify method differences on matched cells with corrected significance testing and effect sizes for each endpoint.

3. **Operational heterogeneity analysis**: we characterize when cost dominance changes by model family, including heavy-tail latency behavior.

4. **Decision guidance**: we derive deployment recommendations that map system constraints to explainer choice.

2 Background

This section summarizes the two explainability methods compared in our benchmark and highlights the technical properties that directly motivate the evaluation metrics and hypotheses used in this paper.

2.1 LIME (Local Interpretable Model-Agnostic Explanations)

LIME approximates a black-box model f locally around an instance x by training an interpretable surrogate model g (e.g., linear regression) on perturbed samples weighted by a kernel π_x. It solves:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g).$$

where $\Omega(g)$ penalizes complexity (sparsity) (Ribeiro et al., 2016). In practice, LIME behavior is strongly shaped by its neighborhood construction and surrogate-design hyperparameters (e.g., kernel width, number of perturbation samples, and feature budget). These controls

often yield compact explanations and low latency, but they also introduce stochastic variation because local neighborhoods are sampled rather than exhaustively enumerated.

2.2 SHAP (Shapley Additive Explanations)

SHAP attributes prediction output to features based on their marginal contribution across all possible coalitions, satisfying axioms of efficiency, symmetry, and dummy properties (Lundberg and Lee, 2017). KernelSHAP estimates these quantities with a weighted regression scheme over coalitions for model-agnostic usage, whereas tree-specific implementations can exploit model structure. Relative to local-surrogate methods, SHAP is typically denser and computationally heavier in non-tree settings because many coalition evaluations are required to estimate stable attributions.

2.3 Implications for This Benchmark

For a fair LIME-versus-SHAP comparison, the benchmark must control confounders unrelated to explainer design. This motivates the protocol choices implemented in Section 3: identical transformed feature space, shared model artifacts, fixed class target, and matched paired cells by $(\text{model}, \text{seed}, N)$. Under these controls, expected methodological trade-offs become testable: LIME should favor sparsity and cost, while SHAP should favor stability and faithfulness-oriented quality. This framing also aligns with recent calls for multi-metric, functionally grounded XAI evaluation rather than single-endpoint reporting (Pawlicki et al., 2024; Canha et al., 2025).

3 Methodology

This study uses a configuration-driven experimental pipeline for reproducible XAI benchmarking. The pipeline fixes (i) preprocessing and model artifacts before explanation, (ii) explainer hyperparameters via versioned YAML configurations with deterministic seeds, (iii) artifact-level logging per run (including `results.json`), and (iv) explicit validity filters plus matched SHAP-LIME pairing for inferential analysis. This structure follows recent recommendations for functionally grounded, multi-metric XAI evaluation and transparent benchmarking (Canha et al., 2025; Pawlicki et al., 2024; Agarwal et al., 2022). We instantiate this pipeline for a focused LIME-versus-SHAP comparison to produce claim-ready evidence on quality-cost trade-offs under controlled conditions.

3.1 Research Questions and Hypotheses

The methodology is model-agnostic and reproducibility-oriented, integrating quantitative endpoints with a secondary semantic assessment layer.

We address the following questions:

1. **RQ1 (Robustness)**: Is SHAP more stable than LIME under input perturbation?

2. **RQ2 (Parsimony)**: Is LIME sparser than SHAP?

3. **RQ3 (Efficiency)**: Is LIME computationally cheaper than SHAP?

4. **RQ4 (Faithfulness)**: Does SHAP provide stronger faithfulness/fidelity signals than LIME?

Corresponding directional hypotheses are:

1. $H_1 : S_{\text{SHAP}} > S_{\text{LIME}}$ (stability).
2. $H_2 : P_{\text{LIME}} > P_{\text{SHAP}}$ (sparsity).
3. $H_3 : C_{\text{LIME}} < C_{\text{SHAP}}$ (cost).
4. $H_4 : F_{\text{SHAP}} > F_{\text{LIME}}$ (fidelity/faithfulness).

3.2 Experimental Design and Analysis Units

The robustness cohort is instantiated from the repository-tracked EXP2-scaled matrix with factors:

- model family $g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}$,
- explainer $k \in \{\text{lime}, \text{shap}\}$,
- seed $s \in \{42, 123, 456, 789, 999\}$,
- sampling intensity $N \in \{50, 100, 200\}$ per TP/TN/FP/FN stratum.

This defines $5 \times 2 \times 5 \times 3 = 150$ planned LIME/SHAP runs. In the current artifact state, we retain 140 analyzable LIME/SHAP runs (75 LIME, 65 SHAP) and 65 matched SHAP-LIME cells on identical (g, s, N) coordinates for paired inference. Accordingly, this manuscript reports the EXP2 robustness cohort rather than the earlier EXP1 calibration subset, because paired multi-model coverage is required for inferential claims.

To avoid pseudoreplication, we use a hierarchical analysis structure:

1. **Instance level**: per-case metric values.
2. **Run level**: mean metric per configuration.
3. **Matched-cell level**: paired SHAP-LIME run means on shared (g, s, N) cells (primary inferential unit).

3.3 Data, Preprocessing, and Model Controls

All experiments use the UCI Adult data set (Kohavi, 1996) with a stratified split and deterministic seeds. The preprocessing pipeline follows train-only fitting with:

- numeric scaling (‘StandardScaler‘),
- categorical one-hot encoding (‘OneHotEncoder‘ with `handle_unknown='ignore'`).

Pre-trained black-box models are loaded from fixed artifacts and reused across explainers to isolate explainer effects: random forest (Breiman, 2001) and gradient-boosted trees (`xgb`) (Chen and Guestrin, 2016), alongside additional tabular families present in the robustness cohort.

Evaluation instances are sampled by error quadrants (TP, TN, FP, FN) via stratified sampling. Nominal run size is $4N$, with realized counts bounded by quadrant availability and runtime failures (observed range: 27 to 800 instances; median 400).

3.4 Explainer Protocol

Both explainers consume the same transformed feature space and target the positive class probability:

- **SHAP**: TreeExplainer for tree models, KernelExplainer for non-tree models, with background subsampling ($n = 50$).
- **LIME**: tabular local surrogate with `num_features=10`, effective `num_samples=1000`, and `kernel_width=3.0` in the EXP2 configuration family.

3.5 Metric Operationalization

Primary metrics are computed per instance and aggregated per run:

1. **Stability** S: mean pairwise cosine similarity of attribution vectors under Gaussian perturbations ($T = 15$, $\sigma = 0.1$):

$$S = \frac{2}{T(T-1)} \sum_{a<b} \cos\left(e^{(a)}, e^{(b)}\right).$$

2. **Sparsity** P: active-feature ratio above threshold $\tau = 10^{-4}$ (lower active ratio indicates sparser explanations).
3. **Fidelity** F: faithfulness correlation between attribution magnitude and prediction drop under feature masking.
4. **Faithfulness Gap** Δ_k: absolute prediction change after masking top-k features ($k = 5$):

$$\Delta_k = \left| f(x) - f(x_{\text{mask-top-k}}) \right|.$$

5. **Cost** C: per-instance wall-clock explanation time (ms).

3.6 Inferential Protocol

The statistical plan follows established comparative-ML practice (Demsšar, 2006) and thesis ADR decisions:

1. primary test: two-sided paired Wilcoxon signed-rank on matched SHAP-LIME cells for each metric (Wilcoxon, 1945),
2. effect size: paired d_z for magnitude reporting (Lakens, 2013),

3. multiplicity control: Holm-Bonferroni over the five primary metrics (Holm, 1979).

All inferential claims are restricted to artifact-qualified runs: parseable, non-empty results with explicit matched pairing.

3.7 Secondary Semantic Evaluation

Aligned with this paper's objective of producing claim-ready comparative evidence on LIME-versus-SHAP quality-cost trade-offs, the framework includes an LLM-based semantic layer (clarity/intuitiveness-style scoring). This layer is auxiliary to the primary functionally grounded endpoints and is not used to test H_1-H_4, compute adjusted p-values, or determine method superiority (Pawlicki et al., 2024; Canha et al., 2025).

Methodologically, we position semantic scoring within the current LLM-as-a-judge evidence base: rubric-guided judging can be scalable and informative, but requires explicit reliability framing because outputs are sensitive to judge bias and prompt design (Zheng et al., 2023; Gu et al., 2024). Domain-specific evidence also shows that rationale-aware LLM judging can correlate strongly with expert assessment when the rubric is structured (Zhou et al., 2025).

At the same time, proxy LLM judges may diverge from end-user perceptions, so semantic scores are interpreted as contextual evidence rather than human-ground-truth utility estimates (Wu et al., 2025). Therefore, inferential conclusions in this paper remain anchored in technical metrics, with semantic outputs used to support interpretation of practical deployment trade-offs. In this artifact freeze, no human-agreement coefficient (e.g., κ) is claimed for the semantic layer in this paper.

3.8 Reproducibility Contract

The protocol is fully configuration-driven (YAML + seeded execution), with per-run ‘results.json‘ artifacts, explicit exclusion of malformed or empty runs, and no synthetic imputation. This provides a deterministic audit trail from configuration to manuscript-level claim.

4 Results

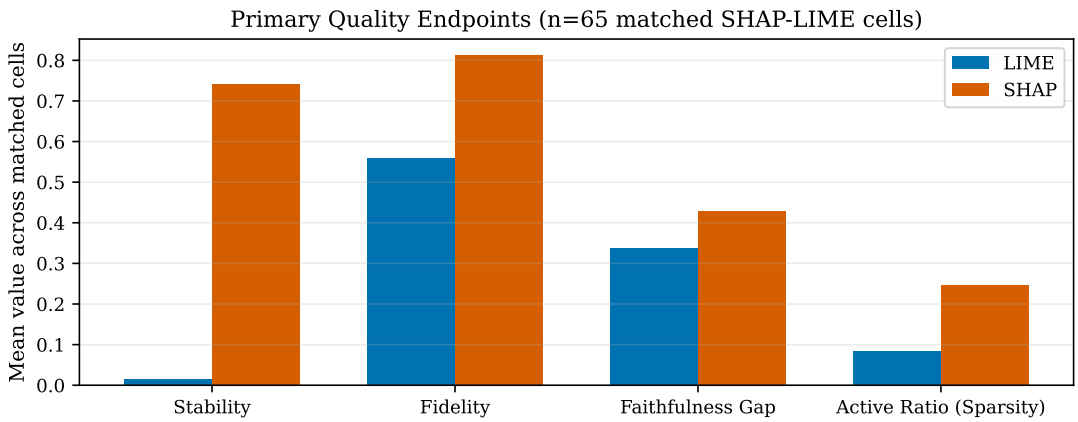
Results are reported from the artifact-qualified EXP2 robustness cohort. Of 250 run files, 233 are analyzable; for this paper's paired LIME-versus-SHAP inference we retain 140 analyzable runs (75 LIME, 65 SHAP) and 65 matched cells with identical $(\text{model}, \text{seed}, N)$. Matched coverage spans all five model families (xgb = 15, logreg = 15, rf = 15, mlp = 11, svm = 9) and all sampling intensities $N \in \{50, 100, 200\}$ with near-balanced counts.

4.1 Primary Paired Inference

Table 1 summarizes paired run-level outcomes on the 65 matched cells. All five primary endpoints remain significant under Holm-adjusted testing. All numeric entries in Table 1 are source-locked to the analysis export `wilcoxon_shap_lime_all_models.csv` under `outputs/analysis/paper_a_exp2_stats/`.

Table 1: Matched SHAP-versus-LIME paired results ($n = 65$ cells). For all metrics, $\Delta =$ SHAP - LIME. Sparsity is measured as active-feature ratio, so larger values indicate denser explanations.

Metric	LIME Mean	SHAP Mean	Δ	p (Wilcoxon)	d_z
Stability	0.014	0.742	+0.728	2.4×10^{-12}	3.07
Sparsity (active ratio)	0.085	0.247	+0.162	2.4×10^{-12}	0.94
Fidelity	0.558	0.812	+0.254	2.4×10^{-12}	3.97
Faithfulness Gap	0.336	0.429	+0.093	2.4×10^{-12}	0.94
Cost (ms)	3083.1	503870.2	+500787.1	6.5×10^{-10}	0.29



Note: lower Active Ratio indicates sparser explanations.

Figure 1: Paired mean comparison across quality-oriented primary endpoints ($n = 65$ matched SHAP-LIME cells). Active ratio is reported directly; lower values imply sparser explanations.

The stability effect is the largest practical separation in the study. Median paired stability difference is +0.864 in favor of SHAP, and the direction is consistent in all 65 matched cells. This indicates that LIME's local surrogate sampling introduces substantial run-to-run volatility under perturbation, whereas SHAP is comparatively invariant.

For sparsity, the sign is also consistent in all 65 matched cells, but interpretation requires care: the metric here is active-feature ratio. SHAP's larger value means denser explanations; thus LIME is systematically sparser and therefore easier to compress for user-facing narratives.

Fidelity and faithfulness gap both favor SHAP in all matched cells. The fidelity margin (+0.254 mean paired difference) and very large paired effect size ($d_z = 3.97$) indicate not only statistical separation but practically meaningful quality gains under this protocol. Figure 1 visualizes these paired mean separations for the four quality-oriented endpoints.

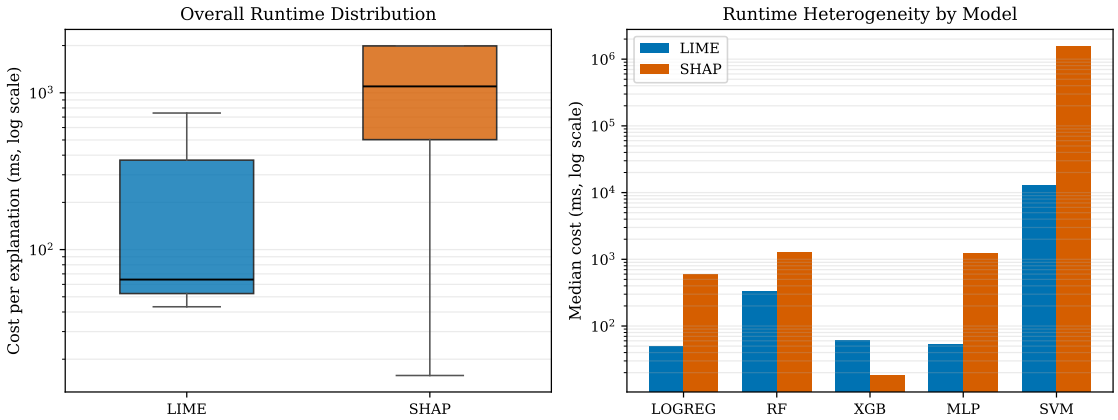


Figure 2: Runtime behavior under matched-cell pairing ($n = 65$). Left: overall cost distribution by method (log scale). Right: model-level median cost (log scale), showing the XGBoost exception where SHAP is faster while other families are SHAP-slower.

4.2 Runtime Distribution and Heterogeneity

The cost signal is significant but strongly heterogeneous. SHAP is slower in 50 of 65 matched cells, with a median cost ratio of approximately 11.5× (SHAP/LIME). However, the distribution is heavy-tailed: for SHAP, the 90th and 95th percentile latencies are approximately 1,539,671 ms and 1,603,727 ms, respectively, compared with 12,825 ms and 13,138 ms for LIME. This tail behavior is concentrated in kernel-SHAP contexts (notably SVM), which inflates mean cost beyond median behavior. Figure 2 shows both the overall log-scale runtime distribution and model-level median heterogeneity.

Heterogeneity analysis by model family shows that quality ordering (stability, sparsity-direction, fidelity, faithfulness gap) is robust, while cost ranking is model dependent. In particular, all 15 XGBoost matched cells show SHAP faster than LIME, whereas RF, SVM, MLP, and LogReg are predominantly or uniformly SHAP-slower. Therefore, "LIME is faster" is valid as a majority statement in this cohort, but it is not universally true across model classes.

4.3 Synthesis of Trade-offs

The empirical frontier is clear: SHAP delivers higher reliability and faithfulness at substantially higher expected runtime, while LIME delivers lower latency and stronger compression at the cost of robustness. The results therefore support objective-driven selection rather than global method supremacy. For assurance-critical analysis, SHAP's quality margin is large and consistent. For interaction-critical systems, LIME's runtime profile remains operationally attractive.

5 Practical Recommendations

Based on these empirical findings, we propose the **Hybrid Deployment Pattern**:

1. **Tier 1: Real-Time / End-User (Use LIME)**
   - Use when explaining predictions to end-users in a live application (e.g., "Why was my loan denied?" in a mobile app).
   - Requirement: < 200 ms latency.
   - Acceptance criterion: explanations are "generally correct" sketches. Reduced fidelity is an acceptable trade-off for usability.
2. **Tier 2: Auditing / Data Science (Use SHAP)**
   - Use when analyzing model bias, debugging model logic, or responding to regulatory inquiries.
   - Requirement: exactness and stability.
   - Acceptance criterion: latency of seconds or minutes is acceptable. Evaluating a batch of data can be done offline.

6 Limitations and Future Work

- **Data Set Scope**: This study focuses on tabular data (Adult). Image and text domains may exhibit different trade-offs.
- **Configuration Space**: We compared standard configurations (KernelSHAP versus standard LIME). Hyperparameter tuning (e.g., LIME kernel width) can alter results.
- **Adversarial Robustness**: We did not subject the methods to active adversarial attacks, only random perturbations.

7 Conclusion

The choice between LIME and SHAP is not a matter of "better," but of "fit for purpose." SHAP is the scientifically rigorous choice for understanding *what the model actually does*, while LIME is the pragmatic choice for *explaining it quickly*. Our paired analysis across 65 matched robustness cells shows consistent SHAP advantages in stability and faithfulness-oriented quality, alongside strong LIME advantages in sparsity and latency for most (but not all) model contexts. This evidence supports architecture-level deployment patterns that separate user-facing low-latency explanation paths from high-assurance auditing workflows.

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